

Protecting Salem's Ecosystem: Predictive Modeling of Stream Temperatures for the City of Salem

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Abstract

Rising stream temperatures are threatening Salem's ecosystem due to many environmental factors. Lower dissolved oxygen levels caused by rising stream temperatures threaten native species, driving them out of local waterways. Additionally, warmer streams create ideal conditions for algal blooms that compromise water quality and surrounding communities. Because tree canopy loss allows direct sunlight to warm waterways, urban restoration efforts frequently focus on tree planting. Here, we aimed to model the change in Salem's Mill Creek temperature from 2015 to 2025 using historical stream temperature, weather, and canopy coverage data with iterative linear regression models. One model stood out among the rest, with an adjusted R-squared value of 0.9442. This model combines interactions between temperature and time to predict stream temperatures. The main driver of the model is time. This means that, as it stands, more work should be done in protecting stream temperature against future rising temperatures, regardless of tree canopy coverage. A defining challenge in this project included limited data availability; thus, it is also recommended that the City of Salem allocate more resources to collecting data that would shed more light on factors impacting stream temperature. Ultimately, this model provides a tool for the City of Salem to build upon, helping policymakers identify which streams are most vulnerable to warming so they can target preventative measures effectively.

Introduction and Background

Bodies of water around the world are beginning to heat up at faster rates than air heat waves (Bush, 2025). These aquatic heatwaves—defined by at least five consecutive days above the seasonal average—result in lower dissolved oxygen content, nutrient cycling, and pH

buffering (PennState, 2026). Sufficient dissolved oxygen is critical for aquatic life and reproduction in streams (EPA, 2025). Lower dissolved oxygen can also impact human use of waterways by inhibiting nitrification, which increases ammonia levels and can cause skin and internal health complications in higher doses (CDC, 2026). Increased stream temperatures also create ideal conditions for algal blooms to develop, increasing both their magnitude and duration (EPA, 2026). These worsening algal blooms threaten humans and animals by producing toxins or releasing harmful gases that sicken those who consume or interact with the water (CDC, 2026). Ultimately, rising temperatures threaten not only native species and cold-water ecosystems, but also human health, recreation, and local water security.

One way to combat high stream temperatures is to support riparian zones, the dense plant areas along rivers (University of Georgia, 2026). These areas often sport trees and other foliage that shade and cool rivers and streams, effectively lowering the water temperature. However, these areas are at risk for several reasons.

The first risk to riparian zones is human development built alongside waterways (University of Georgia, 2026). If projects such as Salem's proposed Riverfront Redevelopment Site are not done explicitly with the ecosystem in mind, there can be negative consequences due to deforestation and disruptions to the riparian zones.

Secondly, Emerald Ash Borers are a new concern in Salem. Emerald Ash Borers are pests that are native to Northern China, Russia, Japan, and Korea (USDA, 2023). These insects are highly destructive to North American Ash Trees and are killing trees across the United States (Penn State Extension, 2025). These beetles were found in Oregon for the first time in 2022 and have been steadily increasing their lifespan as the climate changes (Boboc, 2026). This recent

expansion caused Oregon to start thinking critically about readiness and response plans, including Salem's own plan (City of Salem, 2026).

With these complex system problems in mind, two main research questions were developed with several supporting questions:

1. What features improve future stream predictions from a base model using only stream temperature data, and can a 90% R-squared value be achieved using any combination of features?
2. Can a model be used to predict Salem's stream temperature 3 years into the future based on different levels of anticipated climate change impact on the weather data?

The vision for this project is to provide a preliminary tool for government and private organizations to aid in preventing stream temperature rise in Salem. By developing an effective model that can offer insights into trends in stream temperature and its potential effects on Salem's waterways, this project will seek to define the severity of this issue and provide recommendations to produce further iterations of predictive models. This is a topic that neither the City of Salem nor private researchers have prioritized due to a lack of resources and the recent appearance of the Emerald Ash Borer. This means that there is no previous research on this topic in Salem, Oregon.

This project is of particular interest because the primary investigators were born and raised in the Willamette Valley, surrounded by nature and a curiosity for stewardship. This unification brings a personal element to the heart of the project.

Impact and Stakeholders

This project was outlined with the City of Salem's input and feedback. This topic was already of interest but was also raised by the City of Salem due to their stake in this topic and lack of resources and time to perform the analysis themselves. The results of this project will thus be reported to and will benefit the City of Salem. Using these results, the City of Salem can create a more informed plan to protect riparian zones in Salem, along with the indigenous species that live in them.

The shared theme domain for this project is the Pacific Northwest forest and environmental health. The peer project owners are investigating fire patterns in Oregon. Due to this, the peer project owners could provide different perspectives on topics such as tree weaknesses from drought or heat, as well as heat-related ecological damage in general that may compound this data. This additional perspective will allow the project to consider a more holistic view of rising stream temperatures, their causes, and impacts.

Data and Data Engineering

Data Collection and Cleaning

CastawayWU collected the first data source utilized for the analysis as part of the Mill Creek Research Project. This dataset contains daily average temperatures, dissolved oxygen readings, and turbidity for two sites along the Mill Creek. The two data collection sites were MIC3, in downtown Salem, and MIC12, located southwest of downtown Salem (Figure 1). This dataset was sent to us as a comma-delimited file (CSV) with 7543 rows and 5 columns, and had a file size of 426 KB. The raw dataset is available by request to Dr. Gore or CastawayWU.

Stream Site Locations MIC3 and MIC12

Salem, Oregon

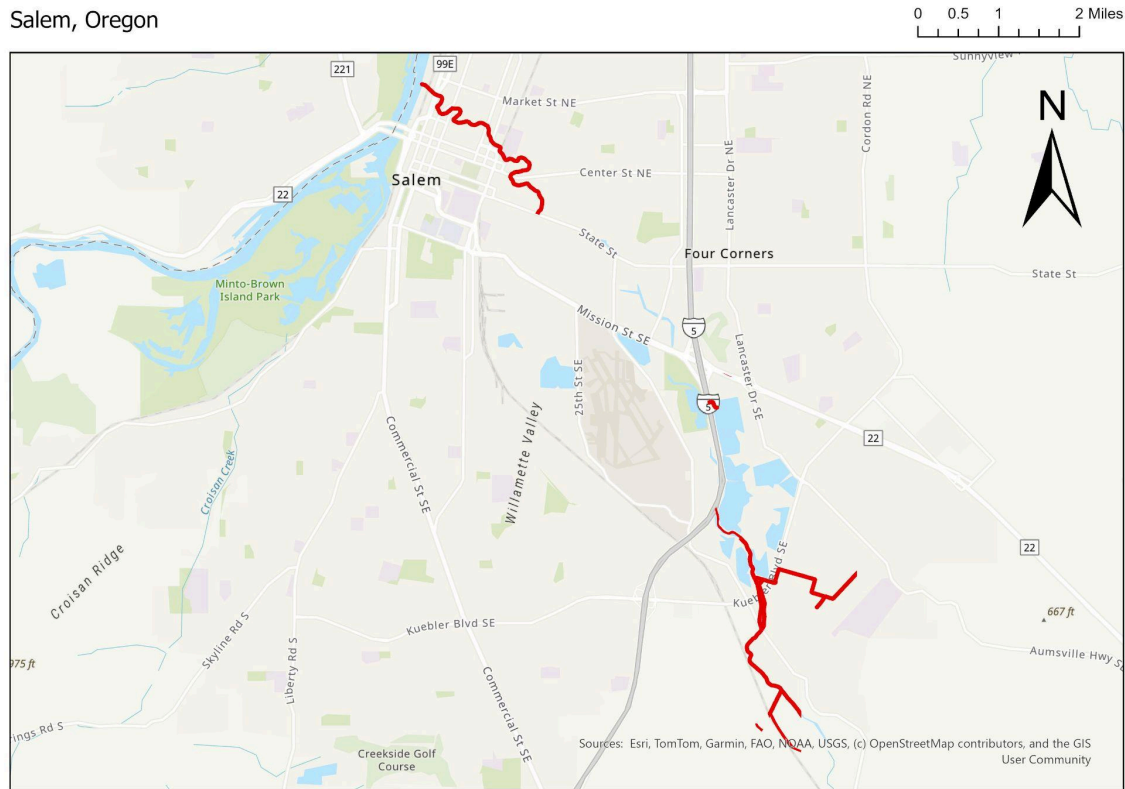


Figure 1: Map of the two stream site locations used in this project. These are the locations from which the stream temperature data and the tree canopy data were pulled. MIC3 is further north and has more tree canopy, whereas MIC12 is further south and gets more sun. Both sites are part of Mill Creek, flowing west into the Willamette River.

Before this stream data could be used in prediction models, data cleaning was completed in an R Markdown document to get the most reliable results and add feature engineering. The first modification to the dataset was to remove the rows of the data before January 1st of 2015. This was necessary because there were only two rows of data from 2014, which would not have supported our models. Then, the day of the year, year, day of the month, and month were added as columns to the dataset to give the models more features to utilize.

The second piece of data that we brought into our project was climate data for Salem from NOAA. This dataset gave us daily information about the location of data collection, the average wind speed, precipitation, if it snowed, snow depth, highest temperature, lowest

temperature, and average temperature. This dataset was downloaded as a CSV file with 48,603 rows and 18 columns, and had a file size of 15.1 MB.

To clean this dataset for analysis, the main task was to retain only the columns that were relevant to our research. These columns provide the information previously cited and have the column names: STATION, NAME, LATITUDE, LONGITUDE, DATE, AWND, PRCP, SNOW, SNWD, TMAX, TMIN, and TAVG. Rows with missing data in TMAX, TMIN, PRCP, and DATE were filtered out of the final dataset.

An important note about this dataset is that null values do not necessarily mean the values should be zero. Zero is entered when there is a record for the different weather recording variables, not null. A majority of these values seem to occur on older dates, rather than newer ones.

The third data source used in this project contained information on public storm creeks and rivers in Salem, authored by the City of Salem. This dataset contained the locations of all bodies of water in Salem, including the two data collection sites along the Mill Creek from the stream temperature data. This data was downloaded as a shapefile with 2,672 rows and 24 columns, and had a file size of 619 KB. This data can be downloaded from DATASALEM.

This dataset contained no missing values or outliers, and no columns were removed. No further data cleaning steps were required in this shapefile for its purposes. This data will only be used to determine the tree density around the two data collection sites but will not be directly used in the models.

The final data source used in the research was tree canopy information from the Tree Plotter Canopy website managed by the City of Salem. This data contained polygons for different levels of tree canopy area across the Salem area. This data was downloaded as a

shapefile with 3,649 rows and 18 columns, and had a file size of 3,184 KB. The map is publicly available; however, a City of Salem login is required to download the raw data. This data is accessible via request from the City of Salem.

The process for generating the final tree canopy dataset uses two raw files: the cleaned creek locations shapefile (`creek_locations.shp`) and the raw tree canopy shapefile downloaded from the Tree Canopy website (`raw_urban_tree_canopy.shp`). Both of these shapefiles were loaded into ArcGIS Pro.

First, the Buffer tool in ArcGIS Pro was used to create a 100ft buffer on the streams to represent the riparian zone. This 100ft buffer represents the geographic areas of focus for tree canopy along streamsides. Next, this buffer layer was used in the Clip tool to clip the tree canopy shapefile, so that the tree canopy data includes only the 100ft streamside riparian zones. Next, the Selection tool was used to select only the stream represented by the MIC3 and MIC12 stream temperature measurement sites. The stream segments corresponding to each of these measurement sites are of similar length. From this layer, the attribute table was exported. Additionally, separate attribute tables for MIC3 and MIC12 were exported so that the two creek segments can be modeled separately. All of these files are attached in the repository for easy reference, as each step was conducted in ArcGIS Pro.

After the GIS processing was complete, the MIC3 and MIC12 tables were wrangled in R to form the final datasets for both creek segments. The tables were converted into long format, creating the variable `UrbanTreeCanopy_pct` as the percentage of urban tree canopy observed that year. Unnecessary columns were removed, keeping 9 columns: `Impervious`, `Water`, `Herbaceous`, `BareSoilDr`, `Unsuitable`, `PossiblePl`, `Shrub`, `Year`, and `UrbanTreeCanopy_pct`. This was then used to create median aggregates for each year for all the variables at MIC3 and MIC12 creek

locations, creating two final modeling datasets: mic3_summary_by_year and mic12_summary_by_year. For other visualization purposes, the non-aggregated versions of the datasets were also saved: tree_canopy_mic3.csv and tree_canopy_mic12.csv.

Data Engineering

The data used in this project was contained in static CSV files or shapefiles, which meant that no continuous refreshes or scraping was needed. This decision was made because the stream temperature and tree canopy coverage data were not frequently updated. The weather data from NOAA was the only continuously updated data source. However, the new information would not have been pertinent to our research without updated stream temperature data on the same dates.

The stream temperature, weather, and tree canopy coverage data that was collected was moved into a PostgreSQL server hosted by Railway. This allowed each group member to access the data when needed. As the data was being moved into the database, a normalized relational schema was created to ensure that all data sources were cleaned and could be joined successfully in the analysis steps (Figure 2).

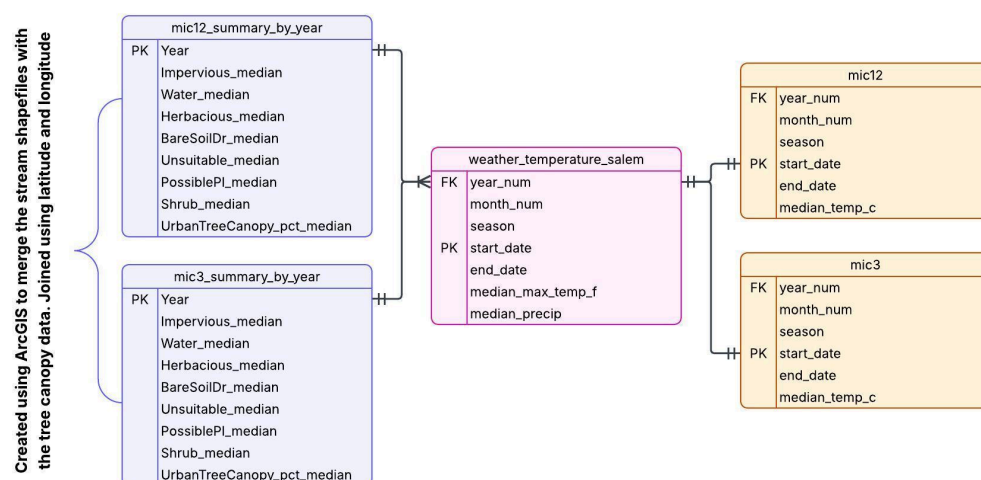


Figure 2: The Entity Relationship Diagram for the normalized database, showing directionality of relationships.

The majority of the analysis for this project was completed in R. Thus, to access the data, commands such as /COPY were used to load the data into CSV files on personal devices. This meant that copies of the original data were stored in each member's file system to be easily accessible for analysis in R.

Ethics and Responsible Use

The main ethical concern in this research is the fit of the model that was created and future creations made with it. Despite using best practices and data sources that were marketed as reliable, the model may generally overestimate or underestimate the change in stream temperature. There tends to be some natural randomness in the weather and stream temperature data, which leads to prediction challenges. In the exploratory data analysis process, it was noted that there were multiple outliers in the stream temperature and weather datasets due to record-breaking temperatures, which might have disproportionately influenced the model towards higher temperatures. To reduce this skew, outliers were removed from the dataset. However, there may have been other similar oddities in the data that were not identified.

While there are ethical concerns regarding this research and its implications, there were no concerns regarding data accessibility and sharing personally identifiable data. Data collected and displayed by the City of Salem is available freely by request, and the CastawaysWU data is also accessible by request. None of our data sources contain information about individuals; thus, sharing this data would not lead to any breach of privacy or confidentiality.

For the most responsible use of this data, users should reference the predictions alongside domain expertise to identify areas to expand preventative measures. The models created in this project should not be used as the one source of truth, despite high accuracy measures.

Methods and Evaluation

To answer the first research question, linear regression models with coefficient interactions were created iteratively in R, with different data included for each model. For the primary question, only the stream data from CastawaysWU was required. Building on the first research question, both the stream temperature and weather datasets were used to determine if these additional factors could improve a model's ability to predict stream temperature. Later, canopy coverage was also added to the model to determine if its factors could improve the model performance. Multiple models were created for each new data source introduced, using different combinations of factors to analyze which columns have the most impact on stream temperature and to create a model that can explain the changes in the data.

Each model was evaluated using the standard graphs given by the `plot()` function in R, along with the R-squared value and RMSE. The residuals vs. fitted graphs were used, as intended, to check the assumption of linearity and ensure that a linear regression fits the data. Then, the normal Q-Q plot was used to check that the residuals created by the model were normally distributed. The Scale-Location graph was used to check the homoscedasticity of the residuals. Finally, the Residuals vs. Leverage plot was used to identify points that are impacting the model more than they should (outliers). In this stage of evaluation, three outliers were identified. Two of these points were recorded on the same day, and the third was the day prior. After additional research, these days were shown to be a part of a “freak, 10,000-year event,” according to author Monica Samayoa at OPB (2022). Thus, these points were removed from the dataset since they were deemed not to be representative of Oregon's weather and stream temperatures. These plots ensured the models being created aligned with the required assumptions for linear regression and were appropriate for use.

While evaluating these models, the primary focus was on creating models with a high R-squared since the goal was a model that would reliably explain variation in stream temperature. However, the RMSE and MAE were also calculated based on the residuals for each model that was created. The goal was to create a model with an R-squared above 0.90 and an RMSE and MAE lower than 5 degrees Fahrenheit. These metrics would signify that the chosen model can explain the variation in the datasets and has low error in its predictions.

Results

Exploratory Data Analysis on Stream Temperature

As previously mentioned, this project analyzed two different stream segments along Mill Creek and used the temperature measurements from MIC3 and MIC12. Figure 3 shows the similarity across temperatures measured at both stream sites. This graph shows that the stream temperature relationship at both sites is 1:1. This means that the temperature changes the same amount at the same time between both sites. Neither site is consistently warmer or colder than the other.

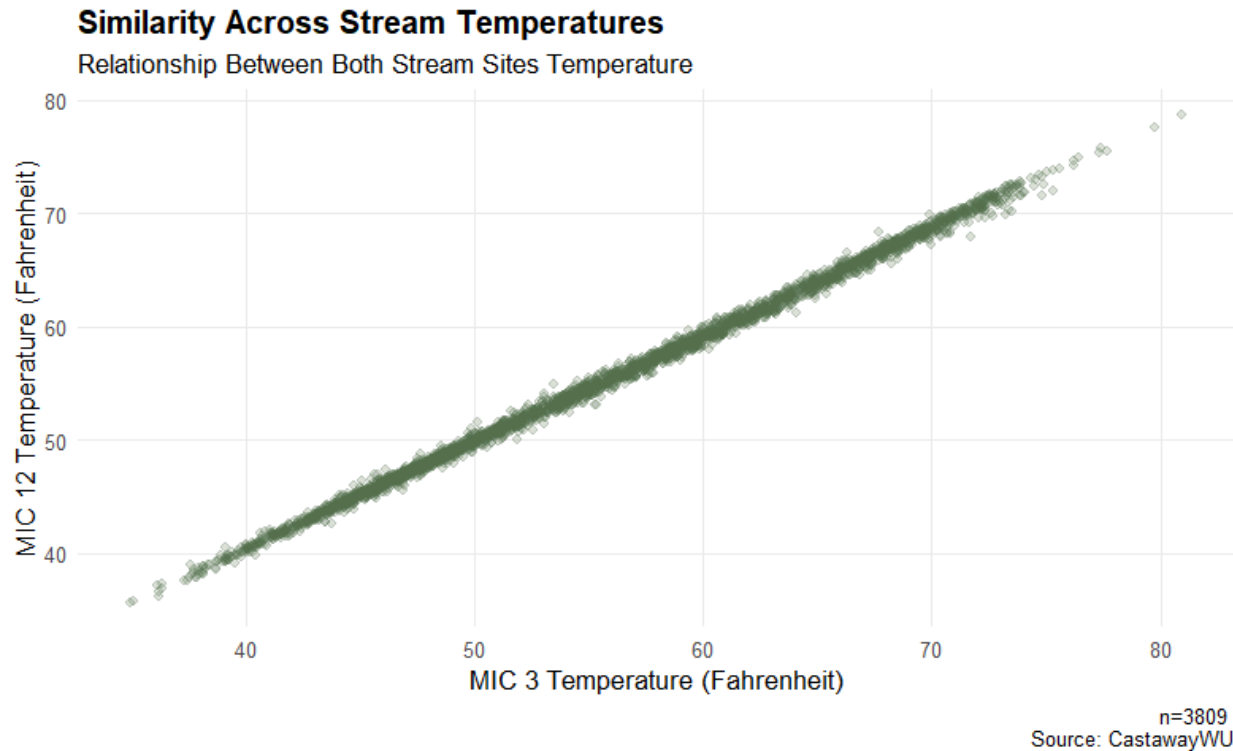


Figure 3: Similarity across stream temperature at both stream sites in Fahrenheit. This data is sourced from CastawayWU and includes 3809 observations between 2015 and 2025. The graph shows a linear 1:1 relationship between the temperatures at both stream locations.

Figure 4 explores the trend in daily stream temperature between 2015 and 2025. The graph shows that even though stream temperatures significantly rise in the summer, there are a few notable outliers. The figure below shows that stream temperatures rose abnormally above yearly trends in 2021 when Oregon hit a record-breaking heatwave in late June. On June 27th and 28th, Salem, Oregon reached a temperature of 117°F, causing stream temperatures in Mill Creek to reach above 80°F.

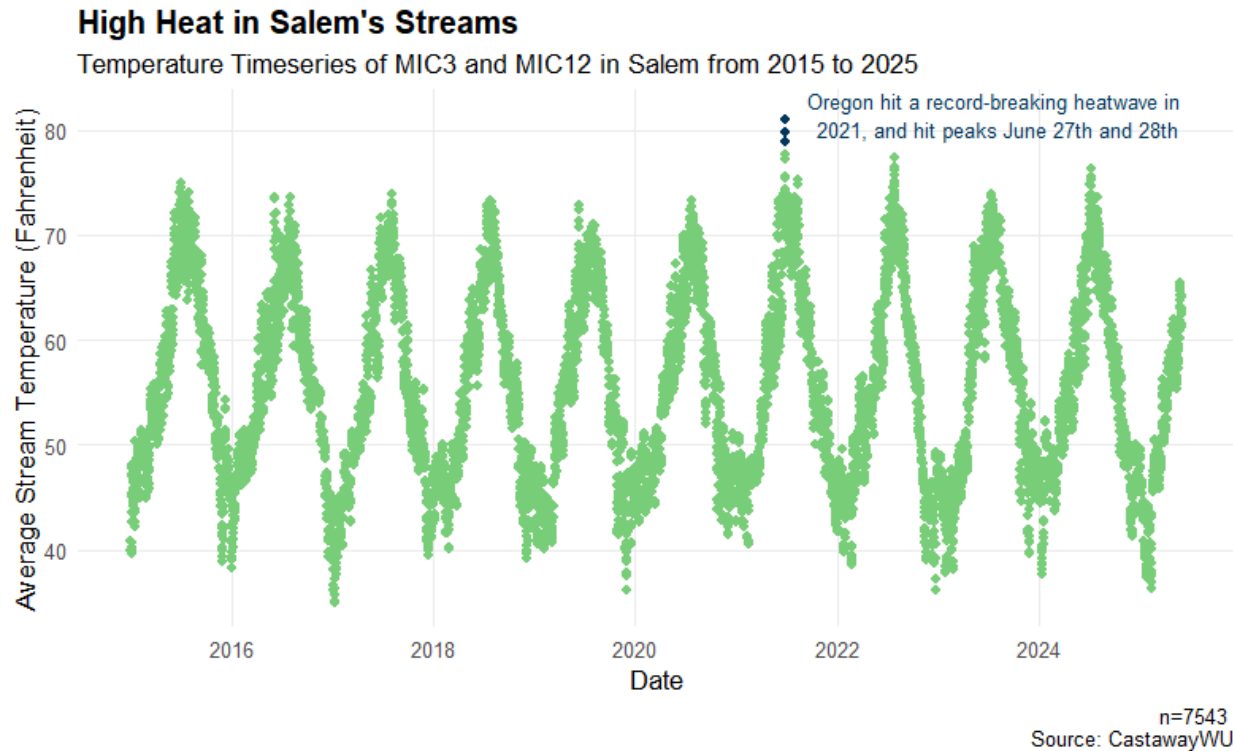


Figure 4: Trends in daily stream temperature between 2015 and 2025. This data is sourced from CastawayWU and includes 7543 observations between 2015 and 2025. The graph shows a sinusoidal trend as temperature changes throughout the year, but it highlights notable outliers in 2021.

Figure 5 explores stream temperature between 2015 and 2025, both daily and seasonally. The graph shows that daily stream temperatures rise and fall each year as seasons change, with warmer stream temperatures in the summer, moderate temperatures in spring and fall, and cooler temperatures in late fall and winter. When looking at stream temperature aggregated by season, we can see a slight upward trend as years progress.

Salem's Stream Temperatures Are On the Rise

Trends in Stream Temperature by Season from 2015 to 2025

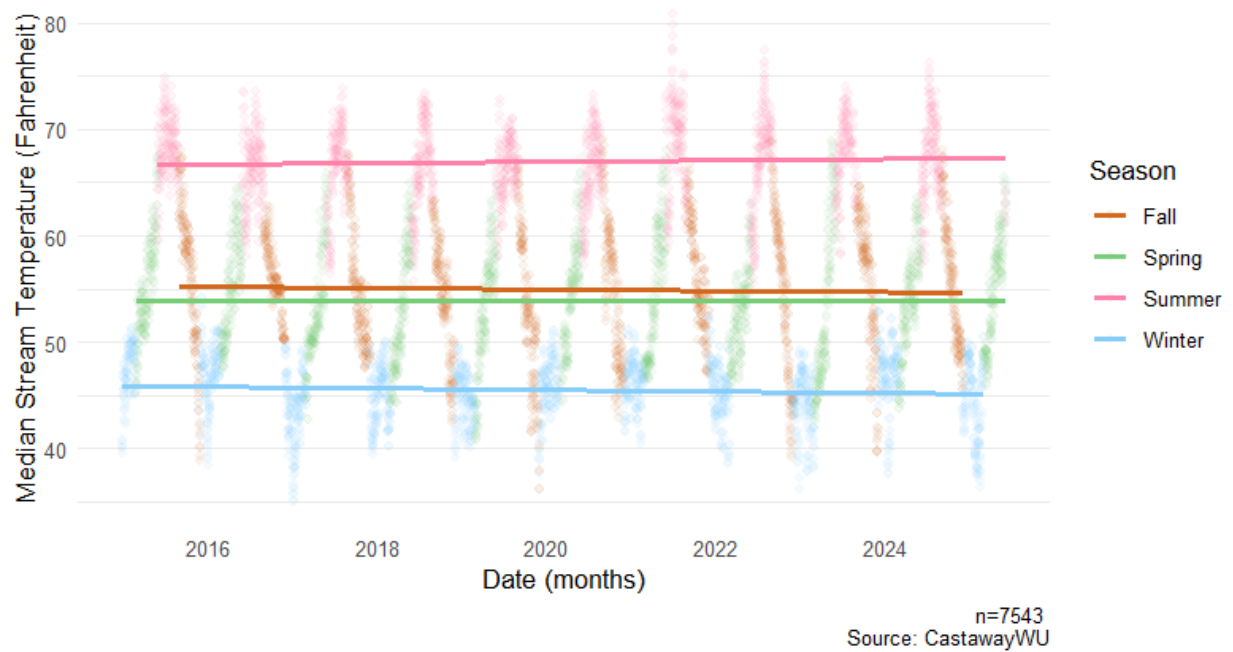


Figure 5: Trends in stream temperature seasonally between 2015 and 2025. This data is sourced from CastawayWU and includes 7543 observations between 2015 and 2025. The graph shows a sinusoidal trend as the temperature changes throughout the yearly seasons. When looking at the aggregations by season, there is a slight upward trend as the years progress.

Figure 6 examines the relationship between stream temperature and dissolved oxygen levels in the water. As stream temperatures rise, dissolved oxygen levels decrease in a clear linear relationship.

Higher Stream Temperatures are Correlated with Lower Dissolved Oxygen

Average Stream Temperature Compared to Average Dissolved Oxygen in the Salem Mill Creek from 2015 to 2025

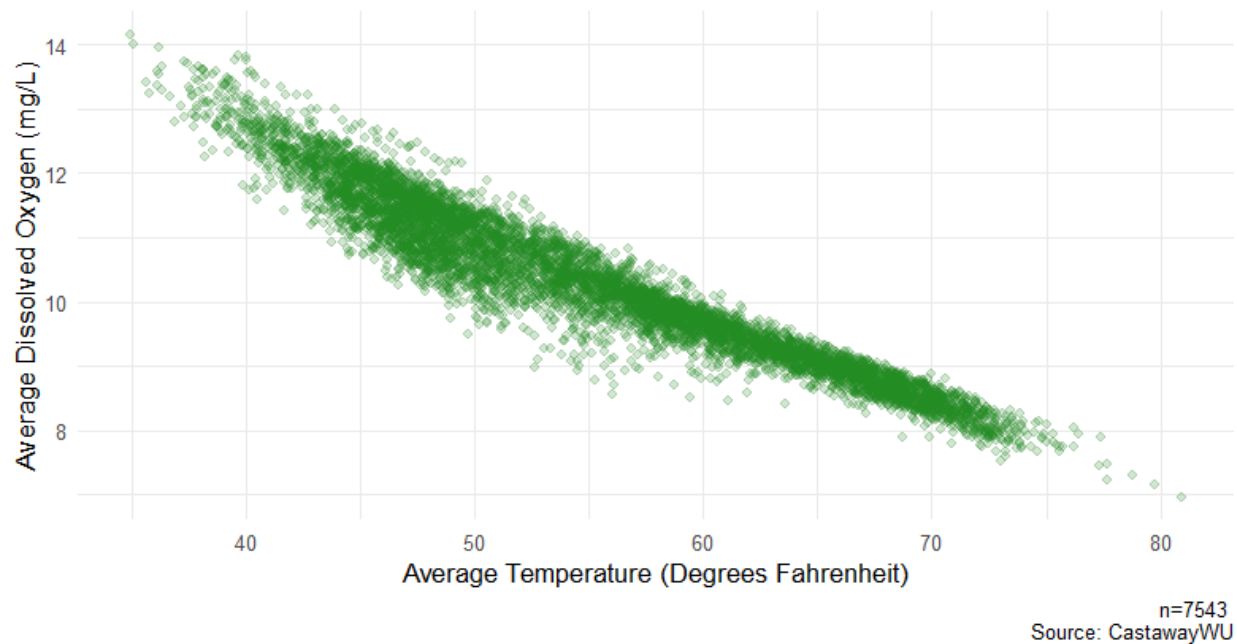


Figure 6: Relationship between stream temperature and dissolved oxygen levels. This data is sourced from CastawayWU and includes 7543 observations between 2015 and 2025. The graph shows a negative linear relationship where dissolved oxygen levels in the water decrease as average stream temperature rises.

Exploratory Data Analysis on Tree Canopy Data

Figure 7 shows each observation for the tree canopy data. This is a total of 6 observations: one observation per year per site for 2020, 2022, and 2024. Most notably, this graph highlights how little tree canopy coverage data there is by year. Figure 7 shows that MIC3 has more tree canopy coverage, ranging between 25% and 33%, whereas MIC1 has less tree canopy coverage, ranging between 5% and 10%. From this limited data, we can see that MIC3

had a slight decrease in tree canopy cover after 2020, whereas MIC12 stayed relatively the same.

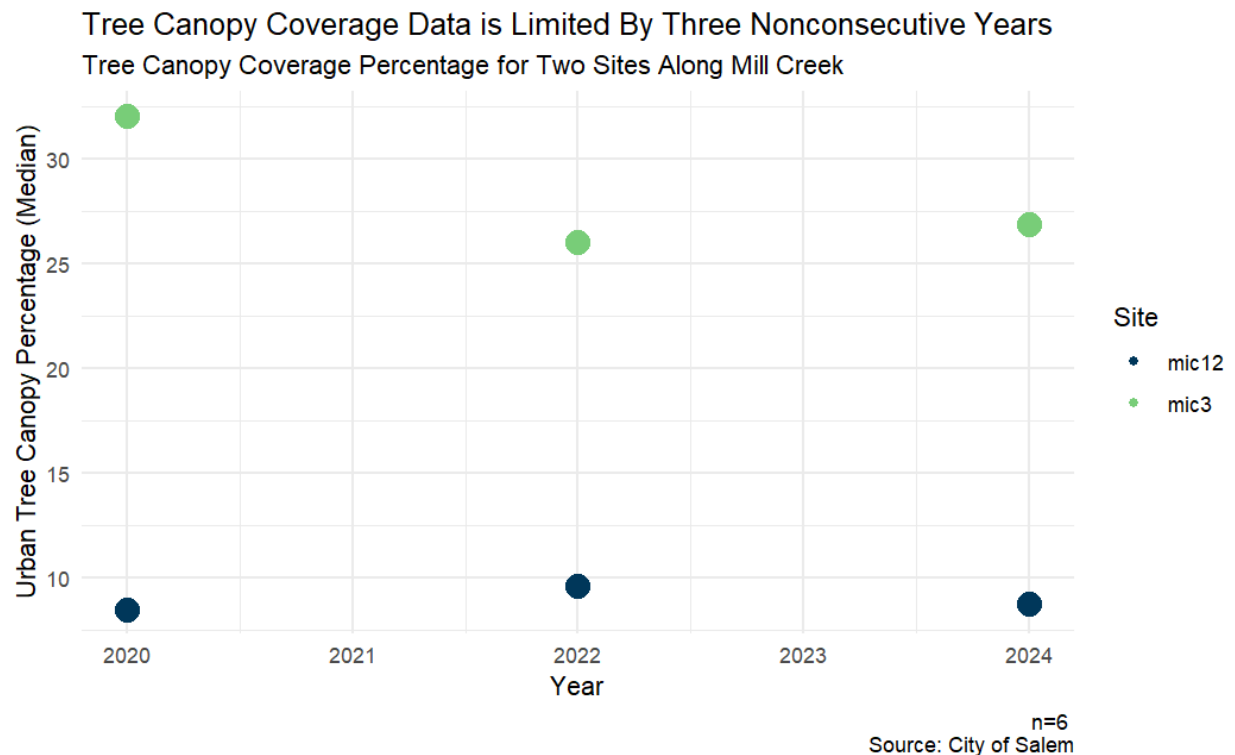


Figure 7: Observations for tree canopy coverage for 2020, 2022, and 2024. This data is sourced from the City of Salem and includes 6 observations. The graph shows the extremely limited nature of the tree canopy cover data, making it difficult to observe year-to-year trends.

Modeling with Stream Temperature Data

The first model used only stream temperature data between 2015 and 2025, and uses year as a character. The only interaction being used is the day of the year squared term. The adjusted R-squared value for Model 1 is 0.8751, and the multiple R-squared is 0.8755. The p-value is less than $2.2e-16$. The residual standard error is 1.755 on 7517 degrees of freedom. The root mean square error is 1.7454. An examination of the diagnostic plot, Figure 8, reveals the existence of 3 prominent outliers. Each plot highlights the same three points, which were determined to be record-high temperature days. These three outliers will be removed as they are not representative of typical stream temperature trends. Outside of the outliers, the diagnostic plots indicate that this data meets the assumptions for a linear regression model.

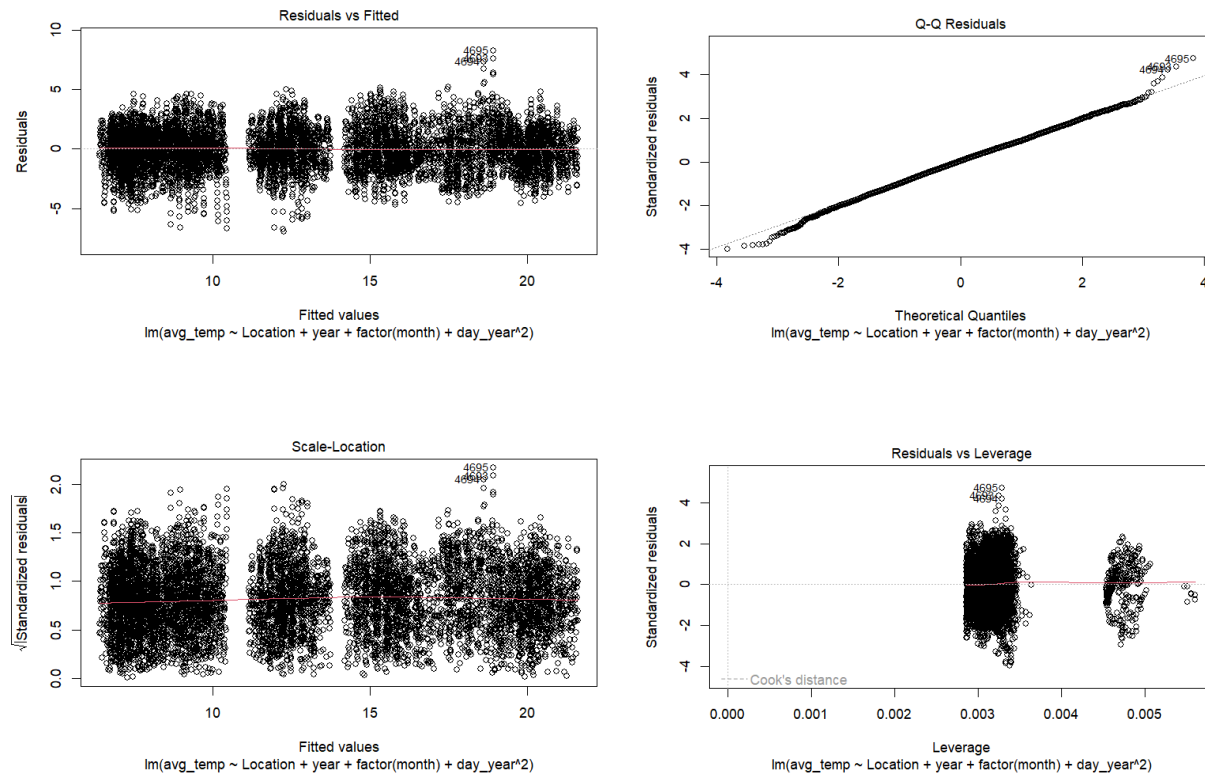


Figure 8: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 1, meaning the assumptions for linear regression are met. Notably, these plots show some outliers—these are a symptom of the 2021 heatwaves in Salem.

The second model uses the same features as Model 1, but without the outliers. The adjusted R-squared value for Model 2 is 0.8757, and the multiple R-squared value is 0.8761. The p-value is less than $2.2e-16$. The residual standard error is 1.748 on 7,514 degrees of freedom. The root mean square error is 1.7454. For this model, examining the diagnostic plots confirms that the removal of the outliers improves the ability of the model to fit the assumptions required for a linear regression model. It is clear that the points highlighted in the graphs as possible outliers are within the typical bounds of stream temperature patterns; thus, no further outliers must be addressed. All of the diagnostic plots continue to confirm that the assumptions for a linear regression model are met.

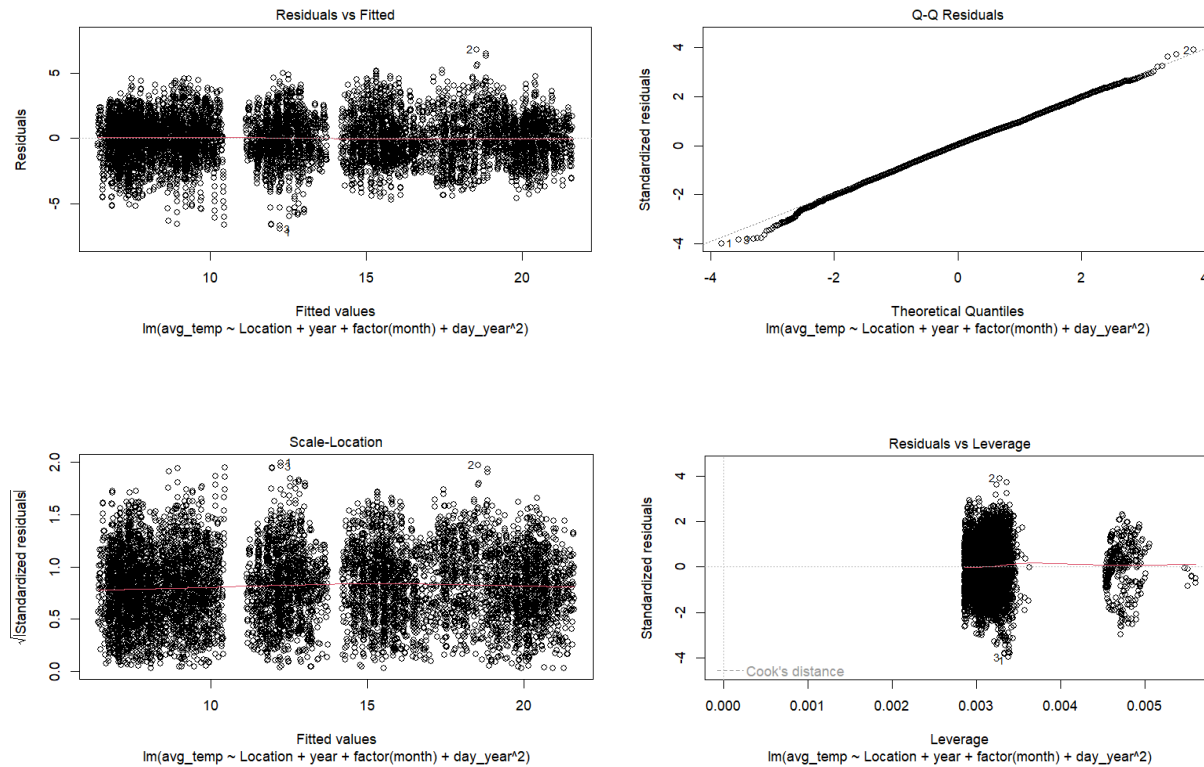


Figure 9: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 2, meaning the assumptions for linear regression are met.

The third model builds off of Model 2 and uses `day_year` as a factor instead of squaring it. The adjusted R-squared value for Model 3 is 0.8997, and the multiple R-squared value is 0.9048. The p-value is less than $2.2\text{e-}16$. The residual standard error is 1.571 on 7151 degrees of freedom. The root mean square error is 1.5298. The diagnostic graphics in Figure 10 confirm that it is not the highest-performing model by showing residual standard errors and additional metrics. This information is visualized as slightly dispersed and clumped points throughout the four graphics shown in Figure 10.

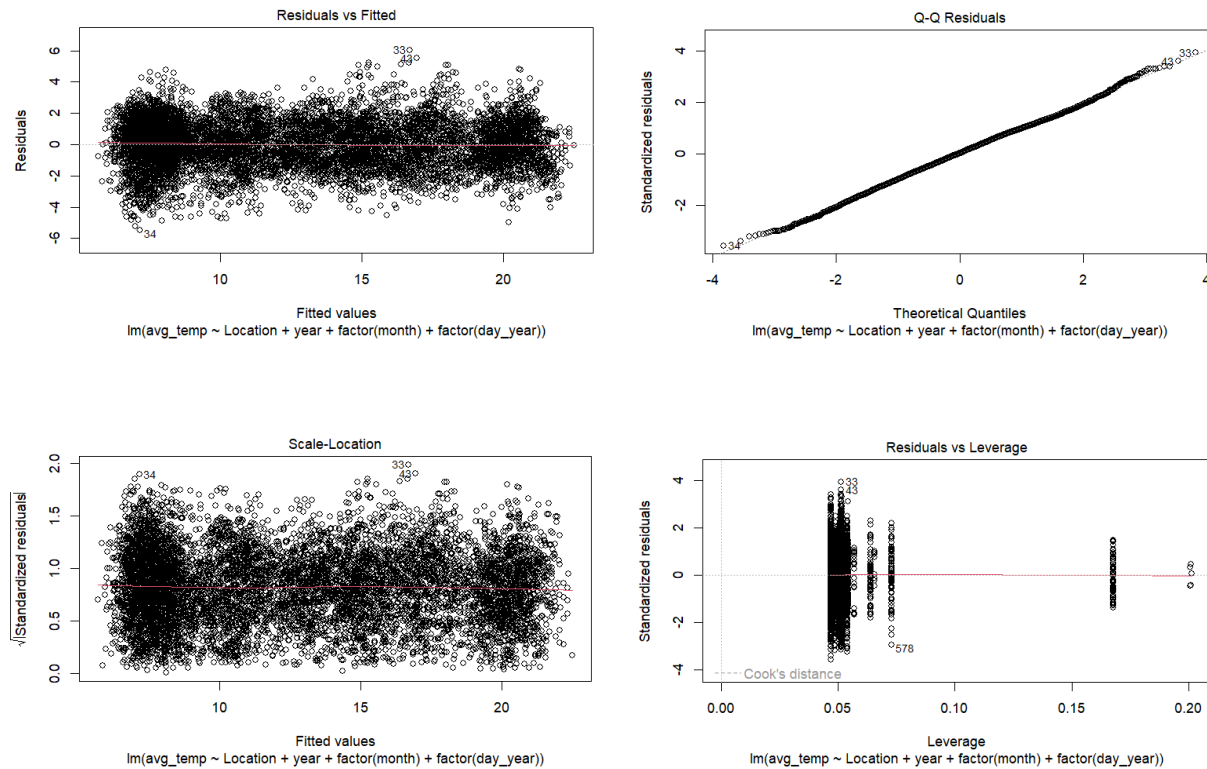


Figure 10: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 3, meaning the assumptions for linear regression are met.

Modeling with Stream Temperature and Weather Data

The fourth model adds weather data variables, uses year as a factor, and includes interactions between variables that have not been used in previous models. The adjusted R-squared value for Model 4 is 0.9442, and the multiple R-squared value is 0.9445. The p-value is less than $2.2\text{e-}16$. The residual standard error is 2.111. The root mean square error is 2.1060. The F-statistic is 3,450 on 37 and 7,503 degrees of freedom. The visualizations of our diagnostic models show stronger results for the dispersion of the residuals, with a clear lack of segmentation and sparsity throughout the graphs.

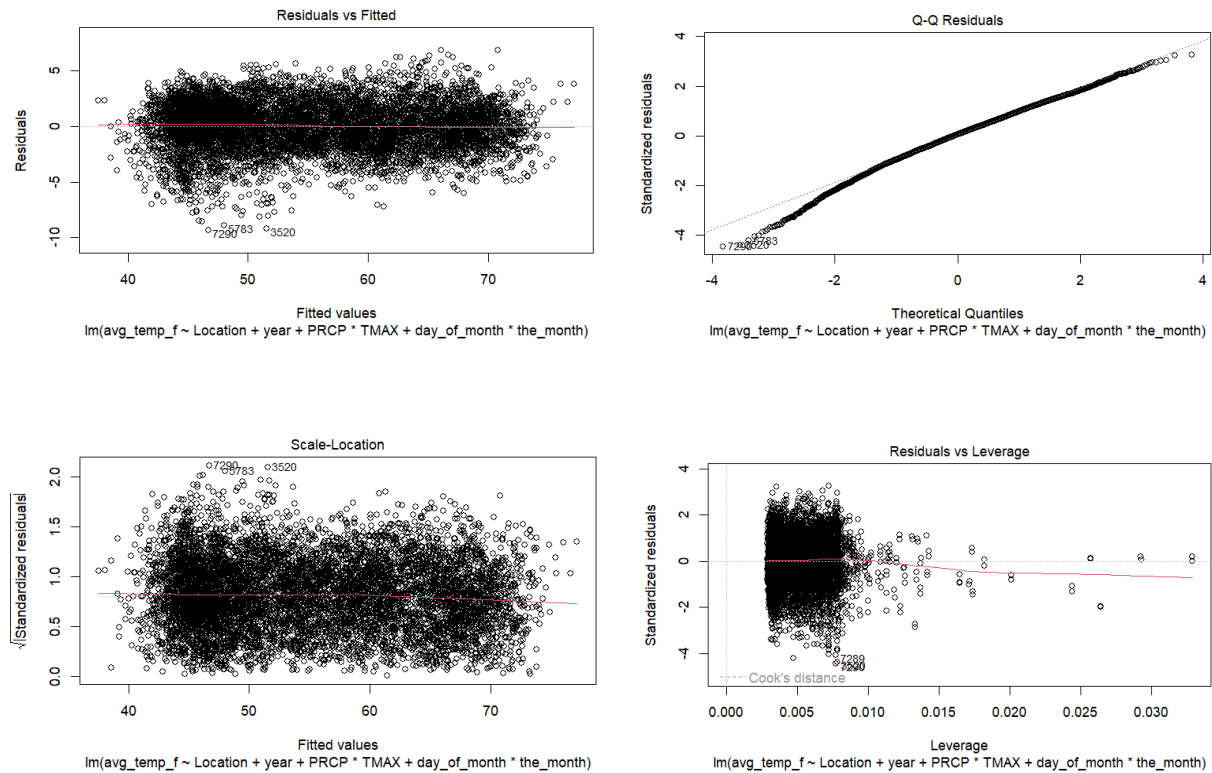


Figure 11: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 4, meaning the assumptions for linear regression are met. This is the best fit of any model in this project, visible here as a lack of segmentation or sparsity in residual plots.

Due to the high performance of Model 4, the fifth model uses its coefficients to predict stream temperature for 2026, 2027, and 2028. However, Model 4 uses a factor for year, meaning that no additional years can be predicted from it. Because of this, an identical model was created using year as a numeric value, with the trade-off being a lower R-squared value but the ability to predict into the future. Model 5 changes year to be numeric and predicts into the future for the next three years. The adjusted R-squared value for Model 5 is 0.9389, and the multiple R-squared value is 0.9391. The p-value is less than $2.2e-16$. The residual standard error is 2.21 on 7,512 degrees of freedom. The root mean square error is 2.2053. The diagnostic graphs for

this model are similar to those shown in Figure 11, but the ones in Figure 12 are slightly less compact, representing the slightly poorer metrics.

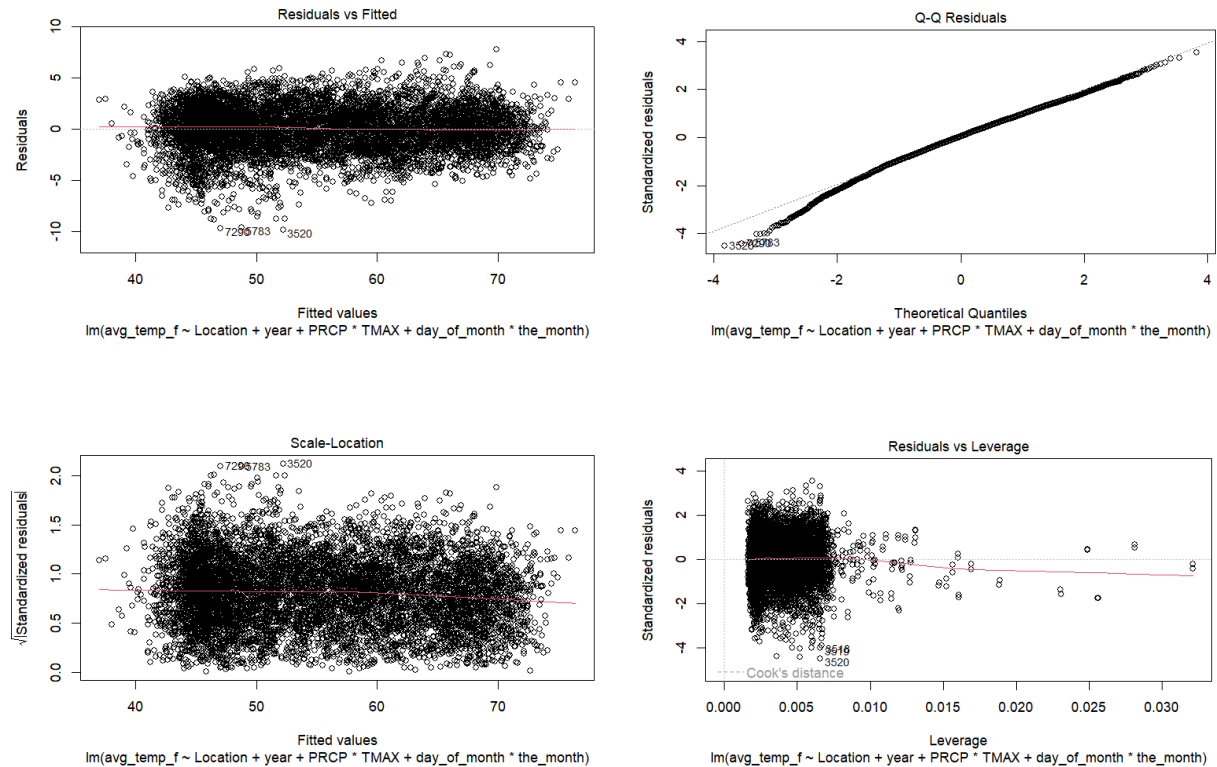


Figure 12: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 5, meaning the assumptions for linear regression are met.

Model 5 allows for predictions into the future using this line of best fit. To test this, a slope was created using 2050's infamous 2.5 degrees of average global warming number to get a by-year linear slope. This is not exact climate modeling, but it will work for testing new, fabricated data in the predictions (Figure 13). This prediction also assumes that all other variables are identical to 2024.

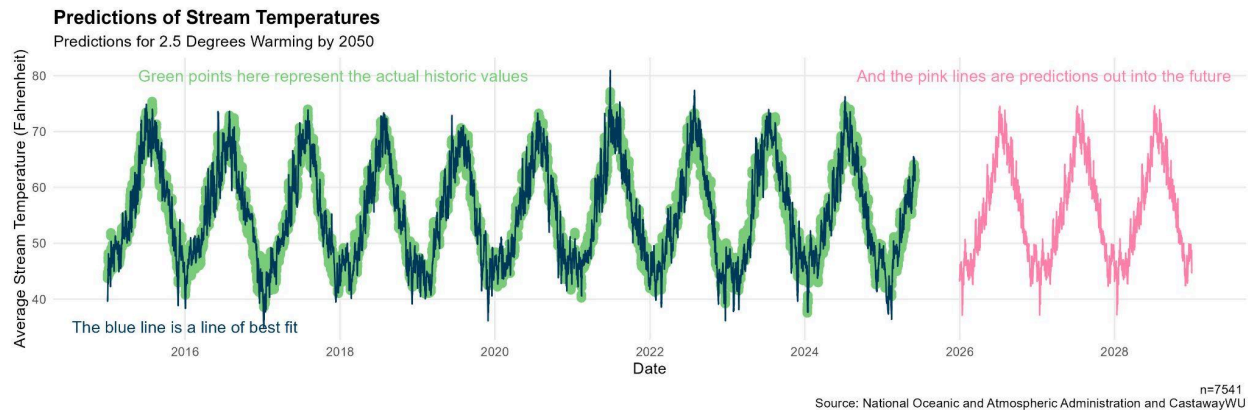


Figure 13: With the line of best fit (blue) overlaid on the actual historic values for stream temperatures (green), this graph displays the predictions (pink) into the future for stream temperature, given all values are identical to 2024 aside from the weather temperature, which is steadily increasing at a slope of 2.5 degrees Celsius of warming by 2050. While this pseudo-future data is not highly accurate, the predictions follow the expected sinusoidal pattern.

Modeling with Stream Temperature, Weather, and Tree Canopy Data

The sixth model incorporates tree canopy data and does not include any interactions aside from day_year^2 . It also does not include additional weather information. The adjusted R-squared value for Model 6 is 0.8811, and the multiple R-squared value is 0.8816. The p-value is less than $2.2\text{e-}16$. The residual standard error is 3.066 on 4300 degrees of freedom. The root mean square error is 3.0595. Figure 14 shows fewer points overall than the previous diagnostic checks, due to the addition of the tree canopy data's limited years. However, these diagnostic visuals show more clustering, representing the worse metrics.

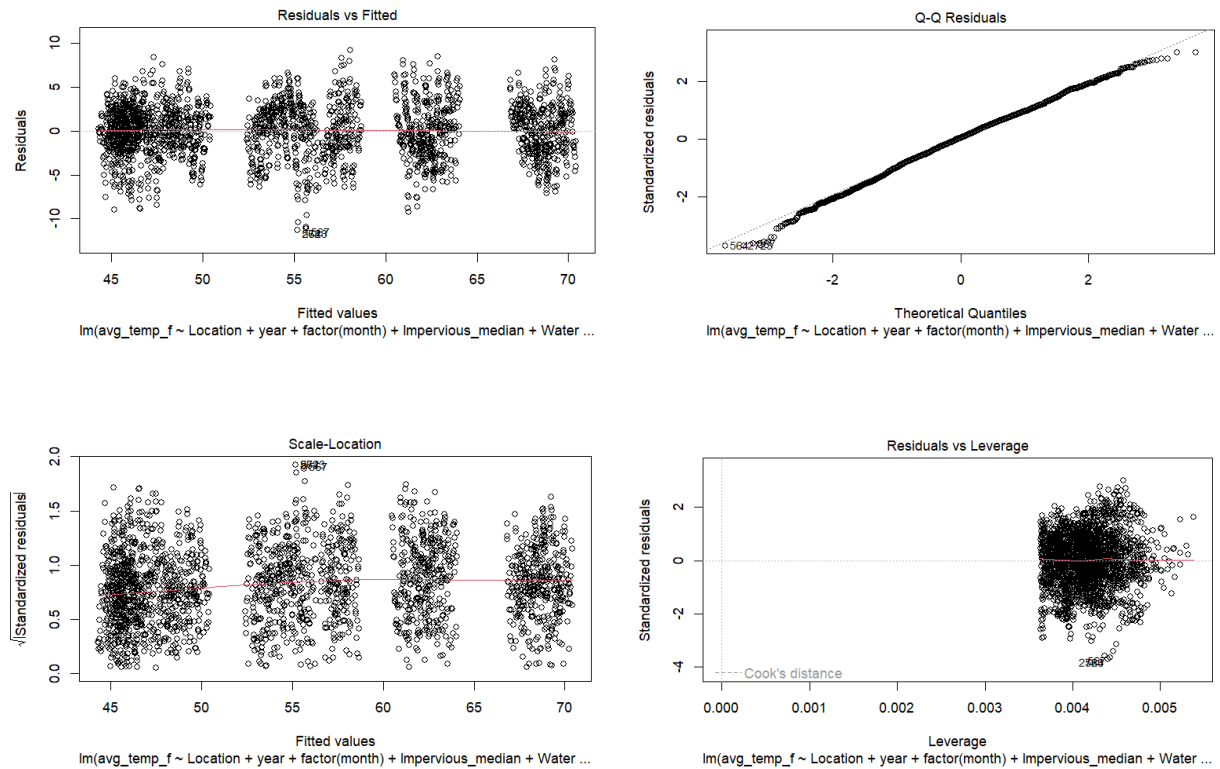


Figure 14: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 6, meaning the assumptions for linear regression are met.

The seventh model is identical to Model 6, but uses a factor for day_year. The adjusted R-squared value for Model 7 is 0.9294, and the multiple R-squared value is 0.9356. The p-value is less than $2.2\text{e-}16$. The residual standard error is 2.362 on 3937 degrees of freedom. The root mean square error is 2.2553. Figure 15 shows more scattered and varied residuals, indicating a poorer fit to the data than diagnostic graphs made without the addition of tree canopy data. However, Figure 15 shows notably less clumped points than Figure 14.

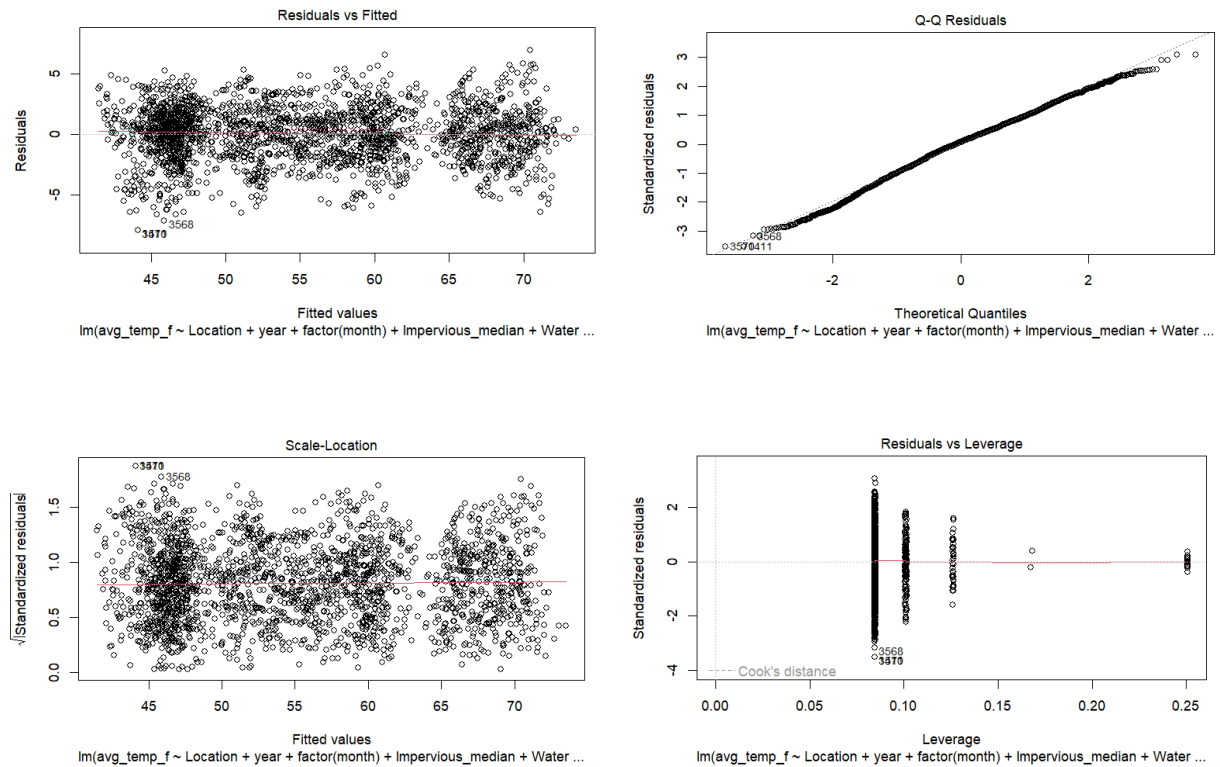


Figure 15: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 6, meaning the assumptions for linear regression are met.

The eighth model builds on Model 7, but includes weather data and uses day_year^2 instead of day_year as a factor with 365 levels. The adjusted R-squared value for Model 8 is 0.9374, and the multiple R-squared value is 0.9337. The p-value is less than $2.2\text{e-}16$. The residual standard error is 2.224 on 4298 degrees of freedom. The root mean square error is 2.2190. Figure 16 shows more uniformly distributed residual points than Figure 14 and Figure 15, indicating a move toward a model with the same data sources and better metrics.

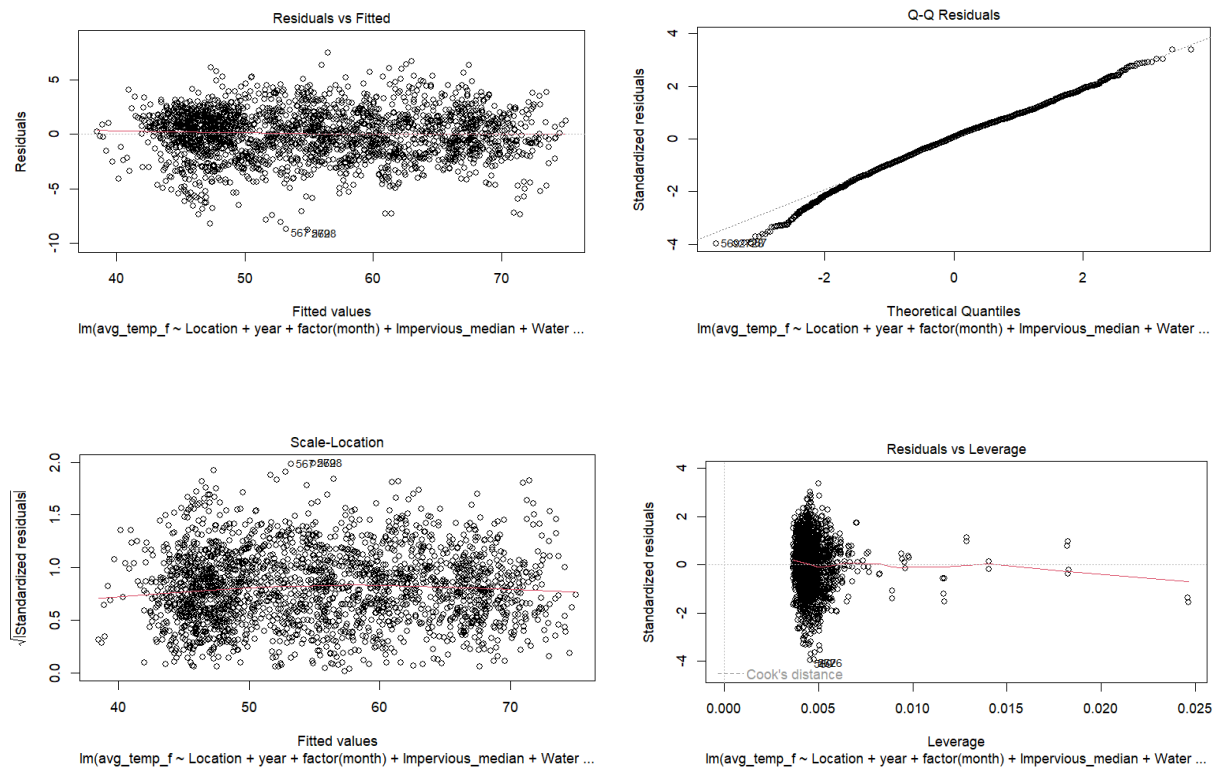


Figure 16: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 8, meaning the assumptions for linear regression are met.

The ninth and final model is identical to Model 8, but incorporates the interactions between the variables. The adjusted R-squared value for Model 9 is 0.9479, and the multiple R-squared value is 0.9482. The p-value is less than $2.2\text{e-}16$. The residual standard error is 2.03 on 4286 degrees of freedom. The root mean square error is 2.023. The diagnostic visualizations shown in Figure 17 contain the most evenly spread residual points, highlighting that this model is the best-performing model. It is important to note that there are also fewer points being used in this residual model, so it is missing the density aspect of the visualization. This means that while the metrics output from Model 9 are the best in the project, they are also built upon less information, potentially contradicting the results.

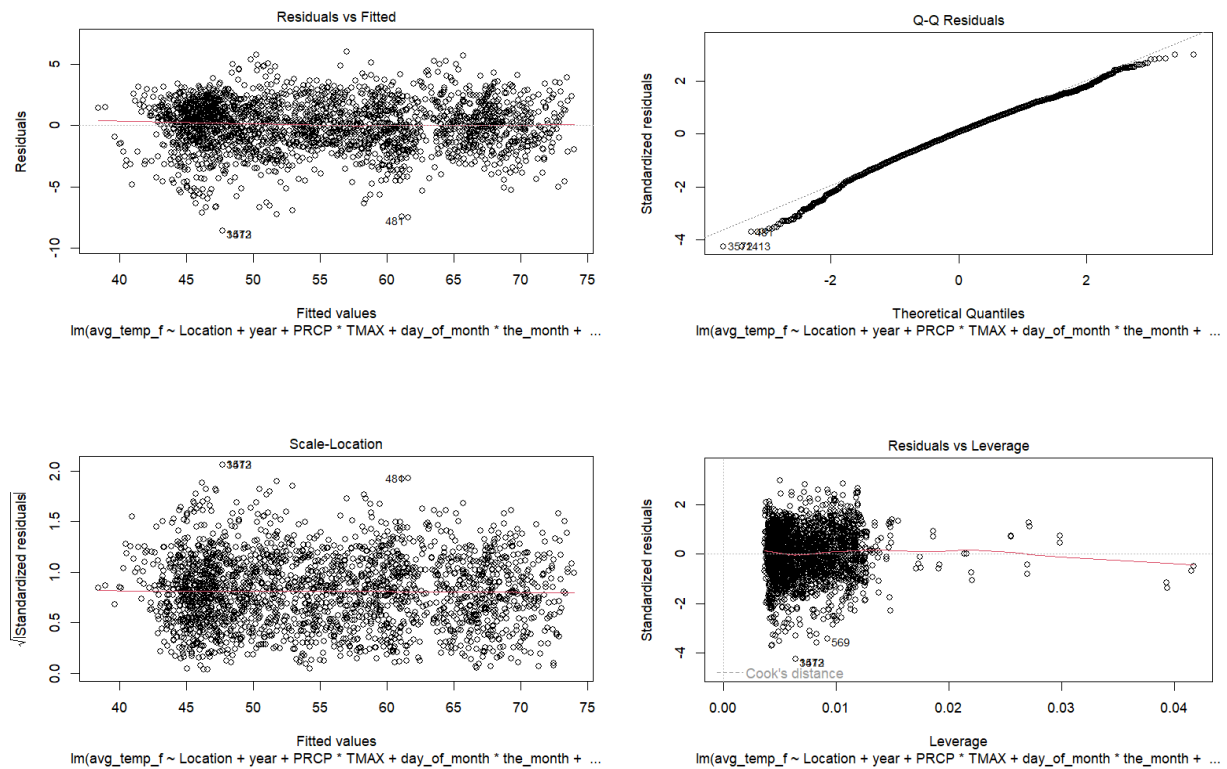


Figure 17: Graphical representations of checked assumptions, including a fitted values vs. residuals plot, a Q-Q plot, a scale-location plot, and a residuals vs. leverage plot. These do not show any unexpected patterns for Model 9, meaning the assumptions for linear regression are met.

Discussion, Limitations, and Future Work

Discussion

This project shows that historic stream temperature data can be used to predict stream temperature with a high R-squared value of at least 90.0%, according to the best model in this project. The inclusion of weather data increases this model's performance with an R-squared value of 94.5%. The addition of tree canopy data increases the model's performance with an R-squared value of 94.7%. However, since this data has an extremely limited range for year, it is difficult to tell if the addition of the tree canopy data would truly improve the model over the

same period as the other models, or if its increased R-squared value is simply due to fewer observations.

Many predictors showed a highly significant relationship with the outcome variable ($p < 0.001$), including location, all years except 2021, precipitation, air temperature, months April through November, the interaction of precipitation and air temperature, and the interaction between day_of_month and the months March through November. See Appendix A for more information about the statistical significance for each variable in Model 4. From this, we can see that the location, weather, and time of year are statistically significant predictors of stream temperature.

Limitations

The first challenge encountered throughout this project was a lack of reliable data relevant to this research topic. Initially, a dataset on the Emerald Ash Borers had been identified that could be used to determine the beetle's impact on Salem's tree population, and this source could be applicable to stream temperatures. This dataset was ruled out after challenges arose when attempting to web-scrape and collect the data. The resulting data from the scrape only included 3 confirmed sightings of Emerald Ash Borers in Salem, which would not have been enough data to add to the models that were created. Another potential dataset contained every recorded tree in Salem, as well as what type of tree it was and where; however, when this data was web-scraped, the results were unreliable. Thus, the project direction had to pivot multiple times as new data sources with different contents were found. This left less time available for in-depth analysis and model creation, resulting in a heftier future work section.

In terms of data limitations, large assumptions were made to create the models. Linear regression models assume that there is no multicollinearity between independent variables.

Unfortunately, multiple variables in the weather dataset are known to be highly correlated, such as precipitation and maximum temperature. To reduce the impact of this multicollinearity, the chosen model, model 4, has the highly correlated variables multiplied by one another in the model fitting. This communicates to the model that it should calculate three coefficients for these two variables to account for the correlation between them. There is still a risk that the model is underrepresenting the significance of highly correlated variables. However, the Variance Inflation Factor value that was calculated indicated that the correlated variables did not negatively impact the chosen model.

Finally, the predictive model has practical limitations which must be considered before its use. The most notable piece to consider is that strong assumptions must be made about future weather to predict stream temperature for that time period. Thus, the results are very dependent on human choices, which can introduce bias. Based on this knowledge, these predictions are meant to give an indefinite sense of possible stream temperatures in the future. Predictions using these models should not be taken as fact, but as a guideline toward action in combination with advice from domain experts.

Future Work

This research could be expanded in many ways to deepen the knowledge on this topic. The first path future research could take would be to include reliable, yearly tree prevalence data in the models. This would provide additional context to the models, which may improve model performance. Similarly, the City of Salem is particularly interested in the impact of Emerald Ash Borer populations on tree loss and on stream temperatures. Once a more robust and accessible dataset containing this information is compiled, it may greatly improve the findings from this research and offer unique insights into the beetle's impact on Salem's ecosystem. Another path

for future work could be to expand the focus to more streams other than Mill Creek. This may help determine Salem-wide trends rather than Mill Creek-specific insights, which may aid in guiding more widespread preventative measures. This premise could be developed further to analyze the area of each data measurement site, as well as the area upstream of that location, since many upstream factors may have great impacts on stream temperature. The final piece of additional data that may aid in the deepened understanding of stream temperature trends would be improved snowfall data. The data used in this study contained only a boolean that indicated if it snowed on that day but did not offer greater details. Each of these additional data sources and approaches may contribute to models with higher R-squared values and lower RMSE.

Finally, future research could explore the use of a seasonal ARIMA model to create predictions of stream temperature. This had been an original part of the plan for this project; however, time constraints shifted focus towards creating a valid linear regression model to predict stream temperature. ARIMA models are known for their applications with seasonal data and forecasting into the future, which may offer greater insights and more reliable predictions.

Conclusions and Recommendations

This study demonstrates that historic stream temperature and weather data can effectively predict stream temperature in Mill Creek, with the best model explaining 94.5% of the variance and an average prediction error (RMSE) of 2.11°F. Air temperature and seasonality were among the strongest predictors. Tree canopy data showed a small improvement in the model, but because only a limited number of years of tree canopy data were available, the current ability is extremely limited. Further research is needed to determine the impact of tree canopy on predicting stream temperature.

Additionally, predictions can be made with the model to predict changes in stream temperature in the next few years. Using pseudo-data, a graph was produced to predict three years into the future. This will be a useful tool to the City of Salem, which has access to more accurate weather predictions that can be fed into these models to predict stream temperatures.

This project should be viewed as an adaptable tool in the early stage of development. Its performance and practicality will grow as more data becomes available. To strengthen this model, it is recommended that the City of Salem invest in collecting consistent yearly tree canopy data and more data for other variables impacting tree canopy, including Emerald Ash Borer infestation and its effect on tree loss. It is also recommended that this model be expanded to include additional stream collection sites beyond Mill Creek to improve the model's ability to represent stream temperature trends across Salem. With these additions, this model has the potential to be a great tool to aid the City in managing stream health, anticipating periods of thermal stress on aquatic life, and guiding decisions for riparian zone and tree canopy restoration.

Reproducibility

For full access to the repository where the project's products—data, markdowns, and other supporting documents—are stored, visit [the Project GitHub](#). Additional information about the process used and project milestones can be found on [the GitHub Project Board](#).

To reproduce this project, Python 3.11.1, R 4.5.2, Beekeeper SQL 5.9, and Railway were used for data modification and storage. Within R and Python, the libraries that were used are listed at the top of the notebooks and markdowns, but are also included here: tidyverse, dplyr, MASS, car, forecast, astsa.

References

Boboc, J. (2026). *Emerald Ash Borer pests are emerging in Oregon at higher rates in May.*

KLCC.

<https://www.klcc.org/environment/2026-05-25/emerald-ash-borer-pests-emerging-in-oregon-at-higher-rates-in-may>

Bush, E. (2025). *U.S. rivers are experiencing unprecedented and unexpectedly intense warming.*

NBC News.

<https://www.nbcnews.com/science/climate-change/us-rivers-are-experiencing-unprecedented-unexpected-intense-warming-rcna233002>

Centers for Disease Control and Prevention. (2026). *Ammonia.*

<https://www.cdc.gov/chemical-emergencies/chemical-fact-sheets/ammonia.html>

Centers for Disease Control and Prevention. (2026). *Harmful Algal Blooms and Your Health.*

<https://www.cdc.gov/harmful-algal-blooms/about/index.html>

City of Salem. (2026). *City of Salem Emerald Ash Borer Readiness and Response Plan.*

<https://www.cityofsalem.net/home/showpublisheddocument/30563/639144417587730000>

Hudgins, E. et al. (2024). *Spread management priorities to limit emerald ash borer (*Agrilus planipennis*) impacts on United States street trees.* Conservation Science and Practice.

<https://conbio.onlinelibrary.wiley.com/doi/full/10.1111/csp2.13087>

Huset, R. (2013). *A GIS-based analysis of the environmental predictors of dispersal of the emerald ash borer in New York.* Syracuse University.

<https://www.proquest.com/openview/a62b54ffa5c9c275cbee2644dfecd010/1?pq-origsite=gscholar&cbl=18750>

Mercker, D. & Hoyt, K. (n.d.). *Forest Stand Delineation: How Is It Done and Why?* University of Tennessee Extension Institute of Agriculture.

<https://utia.tennessee.edu/publications/wp-content/uploads/sites/269/2023/10/W982.pdf>

Oregon Department of Forestry. (2024). *Emerald Ash Borer (EAB) Agrilus planipennis Fairmaire*.

<https://www.oregon.gov/odf/Documents/forestbenefits/fact-sheet-emerald-ash-borer.pdf>

Oregon State University. (n.d.). *Riparian Areas - Stream Side Protection*. Oregon State University.

<https://agsci.oregonstate.edu/water-quality-management/riparian-areas-stream-side-protection#:~:text=Riparian%20zones%20or%20areas%20have,from%20those%20of%20adjacent%20uplands>

Penn State Extension. (2026). *We are in Hot Water – Streams and Rivers are Heating Up*. Pennsylvania State University.

<https://extension.psu.edu/we-are-in-hot-water-streams-and-rivers-are-heating-up>

Penn State Extension. (2025). *Emerald Ash Borer*. Pennsylvania State University.

<https://extension.psu.edu/emerald-ash-borer>

Samayoa, M. (2022). Pacific Northwest heat wave was a freak, 10,000-year event, study finds.

Oregon Public Broadcasting.

<https://www.opb.org/article/2022/09/28/pacific-northwest-heat-wave-2021-oregon-summer-weather-heat-dome-climate-change/>

United States Environmental Protection Agency. (2026). *Dissolved Oxygen*.

<https://www.epa.gov/caddis/dissolved-oxygen>

United States Environmental Protection Agency. (2026). *Climate Change and Freshwater Harmful Algal Blooms*.

<https://www.epa.gov/habs/climate-change-and-freshwater-harmful-algal-blooms>

University of Georgia. (2026). *Stream Warming and Brook Trout in the Southeast*. University of Georgia. <https://storymaps.arcgis.com/stories/834fb9f543424c96ab3f37b677d710fa>

Appendix

Appendix A: Coefficients for Model 4

Variable	Estimate	Standard Estimate	Standard Error	Statistic	P-Value	Significance Code
(Intercept)	33.22313 459	NA	0.250736025	132.502438 1	0	***
LocationMIC3	0.537043 815	0.0300436 47	0.048645662	1.10E+01	4.03E-28	***
year2016	-0.42414 0321	-0.013980 399	0.110967051	-3.8222185 57	0.000133 326	***
year2017	-1.72400 2909	-0.056966 162	0.11110936	-1.55E+01	1.81E-53	***
year2018	-1.65890 0297	-0.054848 588	0.110769979	-1.50E+01	5.52E-50	***
year2019	-1.55461 329	-0.051431 99	0.111518774	-1.39E+01	1.26E-43	***
year2020	-0.75496 1847	-0.024464 733	0.11250108	-6.71E+00	2.08E-11	***
year2021	0.124936 073	0.0041333 18	0.110687626	1.12872664 5	0.259049 253	
year2022	-1.54208 4826	-0.051017 505	0.110806406	-1.39E+01	1.73E-43	***
year2023	-0.56396 2745	-0.018345 802	0.112053389	-5.03E+00	4.94E-07	***
year2024	-0.56581 78	-0.018742 085	0.110669959	-5.11E+00	3.26E-07	***
year2025	-1.22158 5315	-0.027094 2	0.146239446	-8.35E+00	7.83E-17	***
PRCP	-3.51026 4778	-0.097120 502	0.761603758	-4.61E+00	4.11E-06	***
TMAX	0.257003 071	0.4434647 8	0.003346901	76.7883608 7	0	***
day_of_month	-0.01713 1373	-0.016889 294	0.009078568	-1.8870125 46	0.059197 209	.
the_month2	-0.13760 4796	-0.004179 092	0.241363262	-0.5701149 17	0.568616 828	

the_month3	-0.41069 4527	-0.013029 92	0.237924448	-1.7261552 19	0.084360 659	.
the_month4	3.070421 713	0.0967489 53	0.239882691	1.28E+01	4.02E-37	***
the_month5	6.794033 641	0.2180190 12	0.245039605	2.77E+01	3.61E-161	***
the_month6	10.03847 342	0.3055610 7	0.255362907	3.93E+01	1.68E-307	***
the_month7	13.95336 877	0.4288480 37	0.268102751	52.0448549 9	0	***
the_month8	14.67706 514	0.4420186 13	0.28067925	52.2912367 8	0	***
the_month9	9.972864 852	0.3019608 68	0.272309393	3.66E+01	2.61E-270	***
the_month10	6.806545 512	0.2081157 81	0.256241784	2.66E+01	1.15E-148	***
the_month11	3.998212 565	0.1210588 69	0.247092441	1.62E+01	6.51E-58	***
the_month12	0.185719 447	0.0057079 71	0.240731931	0.77147824 3	0.440447 809	
PRCP:TMAX	0.086583 786	0.1324815 14	0.013634119	6.35E+00	2.27E-10	***
day_of_month:the_month2	0.011317 628	0.0057701 26	0.013890459	0.81477706 4	0.415225 792	
day_of_month:the_month3	0.134767 037	0.0799516 34	0.012872465	1.05E+01	1.79E-25	***
day_of_month:the_month4	0.135355 953	0.0763740 74	0.013145568	1.03E+01	1.06E-24	***
day_of_month:the_month5	0.115990 488	0.0690302 67	0.012800188	9.06E+00	1.62E-19	***
day_of_month:the_month6	0.128545 251	0.0698038 07	0.01341341	9.58E+00	1.25E-21	***
day_of_month:the_month7	0.039216 49	0.0223277 82	0.013125635	2.98777837 9	0.002819 226	**
day_of_month:the_month8	-0.10511 4244	-0.058815 586	0.013278456	-7.92E+00	2.80E-15	***
day_of_month:the_month9	-0.08779	-0.047674	0.013599651	-6.46E+00	1.14E-10	***

	4549	387				
day_of_month:the_month10	-0.08540 4961	-0.048226 29	0.013306537	-6.42E+00	1.46E-10	***
day_of_month:the_month11	-0.09097 2734	-0.049400 212	0.013587646	-6.70E+00	2.31E-11	***
day_of_month:the_month12	0.039661 161	0.0225809 53	0.013146356	3.01689374 5	0.002562 355	**

Appendix A: Linear regression coefficients predicting stream temperature for Model 4. These values show the standardized regression coefficients for all of the variables in Model 4. The significance levels include: '***' for a p-value of 0, '**' for a p-value of 0.001, '*' for a p-value of 0.01, '.' for a p-value of 0.05, and blank for a p-value of 0.1.